

21. The Physiology and Pathophysiology of the Kidneys in Aging · assets 1-9 / 10

Box 21.1

Some factors postulated to be involved in renal aging

↓Sirtuin 1/6 (a histone deacetylase enzyme)

↓Klotho expression (and Wnt signaling)

↓Antioxidant production/oxidant activity

↓Energy demand; ↑energy supply (mitochondrial dysfunction)

↓Telomere shortening

↑DNA damage repair

↑DNA methylation

↑Angiotensin II receptor signaling (via Wnt)

↑Cell cycle arrest (G1, via P16ink)

↓Autophagy

↑Fibrosis (transforming growth factor beta [TGF-β]-mediated)

↓Elimination of senescent cells

↓Insulin-like growth factor 1 (IGF-1) signaling

↓Proliferator-activated receptor-γ (PPARγ) activity

↑Capillary rarefaction

↑Podocyte apoptosis or detachment (podocytopenia—absolutely and/or relative to capillary surface area)

↑Vascular sclerosis and glomerular ischemia

↑Advanced glycation end products

↑to-Serine toxicity

↑Endostatin, transglutaminase activation

GI, Gastrointestinal.

Box_21_1

Box 21.2

Kidney and urinary tract diseases that occur more commonly in older adults

Glomerular and vascular diseases

Crescentic glomerulonephritis due to systemic or renal-limited antineutrophil cytoplasmic autoantibody (ANCA)-associated vasculitis

Membranous nephropathy (primary and cancer related)

Monoclonal immunoglobulin deposition diseases (amyloid light-chain amyloidosis and nonamyloid types)

Diabetic nephropathy

Fibrillary glomerulonephritis

Anti-glomerular basement membrane (GBM) disease (mainly females)

Atheroembolic disease

Atheromatous renovascular stenosis

Tubulointerstitial diseases

Acute kidney injury (toxic or ischemic cause)

Lymphomatous infiltration

Renal cell cancer

Urinary tract disorders

Lower urinary tract obstruction due to prostate disease

Ureteric obstruction from cancer (cervical cancer in females)

Transitional cell carcinoma of the bladder

Prostate cancer

Box_21_2

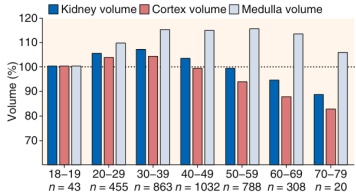


Fig. 21.1 Total kidney, cortical, and medullary volumes among 3509 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Among all donors, older than 40 years, cortical volume decreases and medullary volume increases, making total kidney volume relatively constant until 50 years of age. Beyond that, medullary volume no longer increases, leading to a decrease in total kidney volume from the decreasing cortical volume. Findings were compared proportional to the respective volumes in the youngest age group (18–19 years). (From Wang X, Vrtiska TJ, Avula RT, et al. Age, kidney function, and risk factors associate differently with cortical and medullary volumes of the kidney. *Kidney Int.* 2014;85:677–685.)

Fig_21_1

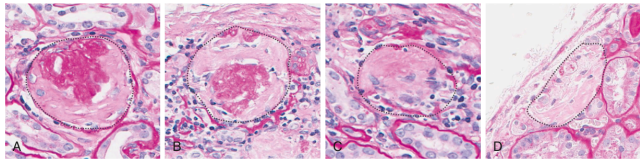


Fig. 21.2 Examples of globally sclerosed glomeruli (GSG) in different stages of their involution. (A and, B) GSG are easily discernible in the biopsy sample. (C and D) GSG that may be overlooked because they progressively atrophy and lose a discernible capsule. All GSG are traced with a black dashed line.

Fig_21_2

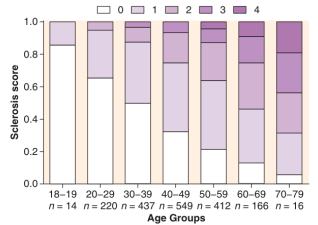


Fig. 21.5 The nephrosclerosis score increases with age among 1814 living kidney donors from the Aging Kidney Anatomy study (expansion of previously reported findings). The total nephrosclerosis score is obtained by adding the individual scores of histologic abnormalities: any global glomerulosclerosis, any arteriosclerosis, interstitial fibrosis greater than 5%, and any tubular atrophy. White bars in all age groups represent a score of 0 (no abnormalities present), and a dark purple bar represents a score of 4 (presence of all four pathologic abnormalities). Three intermediate scores are presented with different shades of purple color. (From Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.)

Fig_21_5

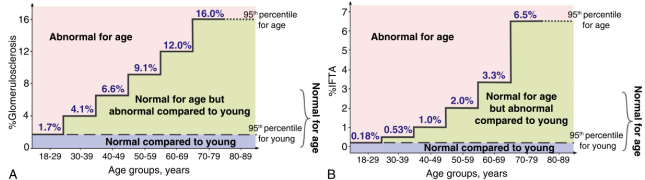


Fig. 21.3 Thresholds for identifying abnormal levels of nephrosclerosis were based on the 95th percentile for 2583 normotensive living kidney donors. (A and B) There were no normotensive kidney donors older than the age of 77 years, so the dotted line in the 80 to 89 years age group was extended using thresholds in donors ages 70 to 77 years. For the nephrosclerosis measures of %Glomerulosclerosis, and % interstitial fibrosis and tubular atrophy (IFTA), the “abnormal compared to young” threshold was 1.7%, and 0.18%, whereas the “abnormal for age” threshold ranged from 1.7% to 16% and 0.18% to 6.5%, respectively. (Modified from Asghar MS, Denic A, Mullan AF, et al. *J Am Soc Nephrol.* 2023;23(8):1421–1432.)

Fig_21_3

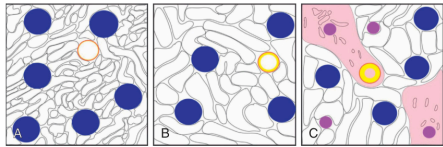


Fig. 21.4 Schematic illustration of how the percentage of glomerulosclerosis influences glomerular density. (A) Example of a young individual with a certain glomerular density (blue solid circle), normal tubules, and normal artery (orange circle), with minimal to no intimal thickening (yellow circle). (B) In an older person in whom the biopsy shows less than 10% glomerulosclerosis, tubules become larger and disperse the glomeruli apart and their density decreases. (C) In older persons in whom there is more than 10% glomerulosclerosis (pink solid circles), the associated interstitial fibrosis and tubular atrophy (light pink areas) bring the glomeruli closer together, thus increasing their density. Arteriosclerosis with intimal thickening (thicker yellow line) is also pronounced in these regions in older adults.

Fig_21_4

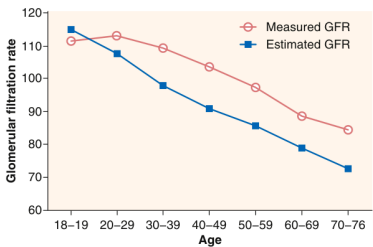


Fig. 21.6 Age-related glomerular filtration rate (GFR; mean values) decline among 4500 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Decline in the estimated GFR (CKD-EPI study equation; see text) is calculated to be 7.4 mL/min/decade and 6.1 mL/min/decade for the measured GFR. (From Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.)

Fig_21_6

Age group, years	Number of glomeruli group (mean) ^a					
	1.0–4.0(4.0)	4.1–8.0(6.3)	8.1–16.0 (12.2)	17.1–32.0(22.8)	32.1–48.0(37.8)	48.1–91.4(59.8)
18–29	0.5	0.5	1.0	1.0	1.0	1.5
30–39	0.5	1.0	1.0	1.5	2.0	2.5
40–49	0.5	1.0	1.5	2.0	2.5	3.5
50–59	1.0	1.5	1.5	3.0	4.0	5.0
60–69	1.0	2.0	2.5	4.0	5.5	6.5
70–77	1.5	2.5	4.0	5.5	7.0	8.5

^aAs predicted by the total number of glomeruli and age from the Aging Kidney Anatomy study.
^bNumber of glomeruli was sum of all complete glomerular profiles (including empty capsules) + .875 × (partial glomerular profiles bisected by the needle).
Adapted from Asghar MS, Denic A, Mullan AF, et al. Age-based versus young-adult thresholds for nephrosclerosis on kidney biopsy and prognostic implications for chronic kidney disease. *J Am Soc Nephrol.* 2023;34(8):1421–1432.

Table_21_1

Table 21.2 Reference values for estimated and measured glomerular filtration rate in potential kidney donors^a

Age group (years)	No.	Estimated glomerular filtration rate (GFR) (measured GFR)			
		Fifth percentile	Median	Mean	95th percentile
18–19	46	88 (87)	114 (107)	115 (111)	145 (144)
20–29	584	81 (84)	108 (111)	108 (113)	130 (149)
30–39	1090	75 (82)	98 (106)	98 (109)	118 (147)
40–49	1364	69 (77)	91 (102)	91 (104)	110 (135)
50–59	1001	65 (72)	85 (94)	85 (97)	103 (133)
60–69	379	61 (64)	78 (87)	79 (88)	97 (118)
70–76	36	55 (57)	71 (85)	73 (84)	91 (113)

^aExpansion of previously reported findings.
From Murata K, Baumann NA, Seeger AK, et al. Relative performance of the MDRD and CKD-EPI equations for estimating glomerular filtration rate among patients with varied clinical presentations. *Clin J Am Soc Nephrol.* 2011;6:1963–1972.

Table_21_2